

Laptop Diagnostics - Hardware Installation and Software Repairment Project

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Tools & Environment: : Lenovo ThinkPad X1 Carbon Gen 4 | Windows 10, Linux Mint (dual boot) | Windows PowerShell | BIOS/UEFI diagnostic utility | Linux Mint bootable USB installer | Windows Event Viewer (disk error logs)

Overview: Configured and diagnosed a Lenovo ThinkPad 4th Gen X1 Carbon that experienced sudden shutdown, immediate battery loss, and driver corruption. The goal of this project is to diagnose the BIOS/ UEFI setup using the Memory Hardware Diagnostics, implementing a dual-boot setup with Linux Mint and Windows 11, and using Windows command-line interface for troubleshooting driver corruption.

Goal: To demonstrate real-world IT troubleshooting and system administration skills by running a full diagnostic on desktop devices, creating & configuring a dual-boot setup with Linux and Windows, repairing driver corruption using Windows utilities, and battery replacements/ repairment.

Procedures Involved

- Created a bootable drive via Rufus, in a dual-boot setup with Windows OS and Linux Mint. After boot-configuration, system and hardware diagnostics were performed to test full functionality of both OSES.
 - During computer diagnostics, discovered power-related issues, such as failing Li-ion main battery and CMOS battery (received a time/ date BIOS error when attempting to access boot menu, indicating a failing CMOS battery)
 - Recently, a C: volume corruption occurred in Event Viewer. Diagnosed and repaired using: 'chkdsk C: /f /r' and 'sfc /scannow'.
 - No further errors/ warnings found
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Laptop #1 - X1Carbon 4th Gen Lenovo Thinkpad Laptop Replacement and Installations



**Outdated 2016 CMOS battery removed, replaced for a newer model.*

**Older laptop battery removed, replaced for new model. The battery chip was damaged prior to battery change, indicating damage caused by the prior owner/ manufacturer. Similar to the CMOS battery, my prior diagnostics related to these components were correct.*

LENOVO Diagnostics UEFI

Time 01:12 - Version 04.

Diagnostic	Status	Progress	Summary
DISPLAY	NOT SELECTED	>	Started at: 2025/09/20 00:31:10
FINGERPRINT	N/A	>	Advanced Integrity Test: PASSED
KEYBOARD	PASSED	100% >	Address Test: PASSED
MOUSE	PASSED	100% >	Bit High Test: PASSED
OPTICAL	N/A	>	Bit Low Test: PASSED
TOUCH	N/A	>	Walking Ones Right Test: PASSED
BATTERY	N/A	>	Walking Ones Left Test: PASSED
CPU	PASSED	100% >	Modulo-20 Test: PASSED
FAN	PASSED	100% >	Moving Inversions-8bit Test: PASSED
MEMORY	IN PROGRESS	76% >	Moving Inversions-32bit Test: PASSED
MOTHERBOARD	PENDING	0% >	Random Pattern Test: Progress [66%]
PCI EXPRESS	PENDING	0% >	Random Number Sequence Test: Waiting
RAID	N/A	>	Block Move Test: Waiting
STORAGE	PENDING	0% >	Bit Fade Test (~100 min): Waiting
WI-FI	N/A	>	Deallocating Memory Step: Waiting

Finished at: 0000/00/00 00:00:00

RESULT CODE:

Total estimated time: 00:53:09 of 04:34:43

FINAL RESULT CODE: 0000000000-00000000

Passed: 04 Failed: 00 N/A: 06 Canceled/Not Selected: 01

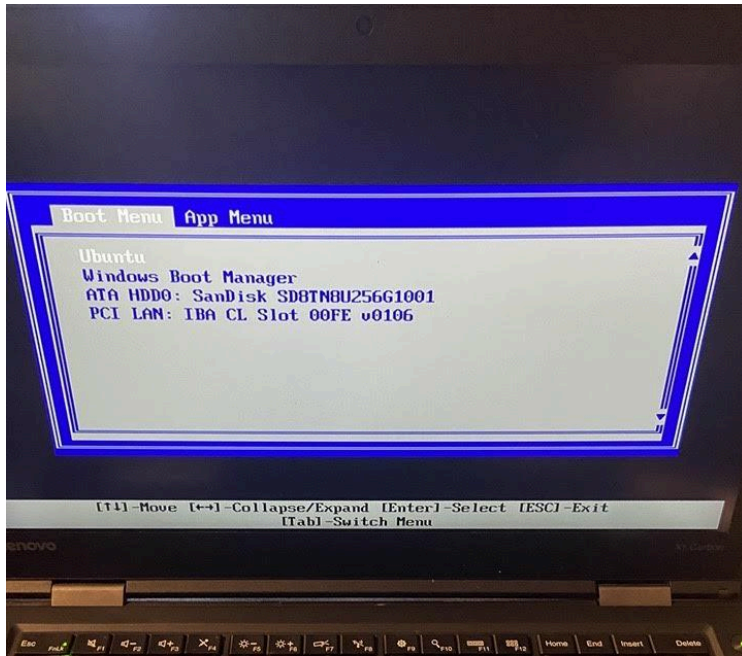
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Navigation [Arrows]

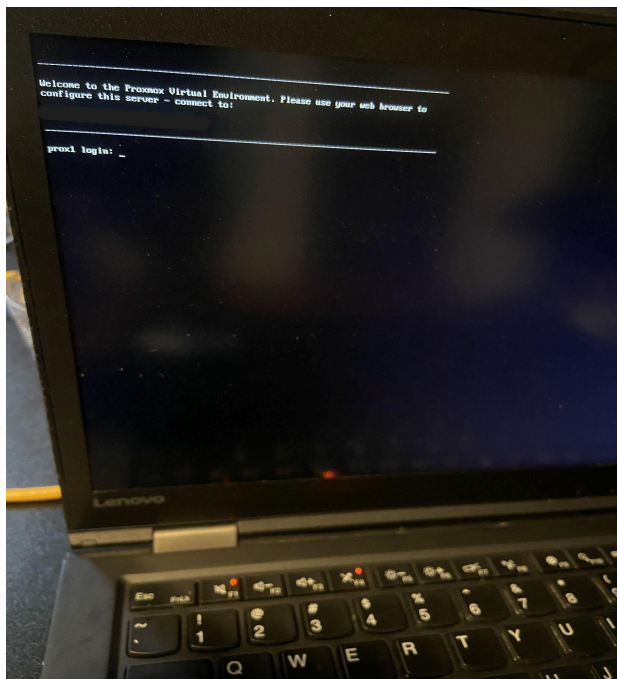
*Lenovo Hardware Diagnostics - 100% Clean

- Checked Battery Report - Laptop Battery decreased by 60%, battery needed immediate change. Charging ports and laptop interior had heavy dust residue, needed deep cleaning inside laptop components and charging ports.
- Windows OS had malware/ lack of OS updating. Rescanned OS using Microsoft Defender, 25% of malware detected from faulty Firefox software installation.
- Deleted and Removed Firefox bug. Currently 100% functional with 0% malware detected according to Microsoft Defender and Antivirus scan

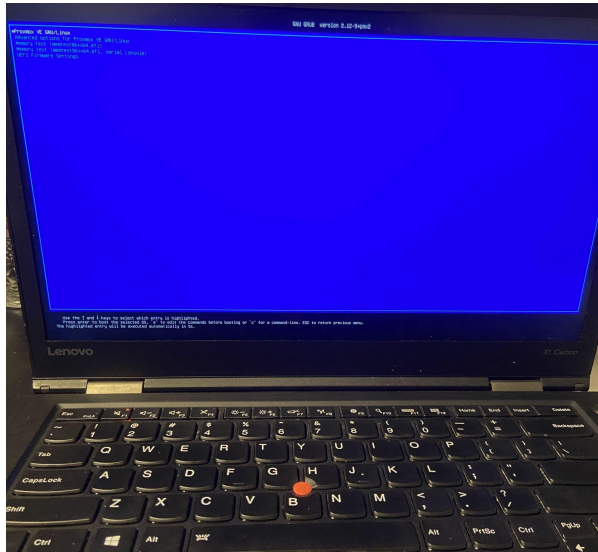
Testing BIOS Settings, Installing New Linux Distribution via Dual Boot Setup



Downloaded Linux Mint into the boot drive, meaning Boot Setup is not corrupted. Disk partitions on Windows and Linux are still intact, and the laptop now serves as an external Proxmox server machine.



Current Status



Lenovo laptop now serves as Proxmox/ OPNsense firewall machine (*Same Proxmox VE Laptop Server used in Project 2: Proxmox Virtualization & OPNsense Firewall Lab*).
Used for educational/ experimental purposes only.

Notes/ Observations

X1 Carbon 4th Gen Lenovo Thinpad Laptop

- The main issues found were boot and power failures caused by CMOS battery depletion and Li-ion battery needs replacement according to Command Prompt: 'powercfg /batteryreport' command via PowerShell.
- Possible cause of driver C: corruption could be poor battery power, sudden shutdowns, or improper dual boot setup. There will be an update on the machine's current condition
- This device was purchased at a local pawn shop, and remained on a shelf with poor maintenance and constant charging. Further diagnostics are needed to be deemed fully functional
- Important Notice: Lenovo Laptop requires total replacement of Li-ion battery and CMOS battery. Hardware BIOS diagnostics scanned 100%, reduced battery power by 30%.

UPDATE STATUS **Laptop is fully operational**

- *New Li-ion battery:* Replaced the old battery with a new one. Decreased time to charge laptop by 85%, increased battery reliability by 90%
- *New CMOS battery:* Time/ Date error during booting stopped at 100%

Laptop #2 - Restoring 2010 Toshiba Windows 7 Laptop

Main Issues

No internet access. Outdated search engine (Internet Explorer) and Norton Antivirus Software slowing down computer

Testing

- Used Search Engine (Google), but there was no internet access.
- The device has not been updated since 2013.
- Lagging persisted, could not even download Firefox browser to substitute Internet Explorer
- Checked Installed Apps on Settings→ Norton Antivirus application outdated, expired membership, and slowed down the entire laptop by 75%.
- Downloaded tool remover to uninstall Norton Antivirus (Windows Vista edition). Outdated removal tools (extensive web searching through open-source forums to find Software Tool Remover for Norton Antivirus - Windows Vista Edition).

Report

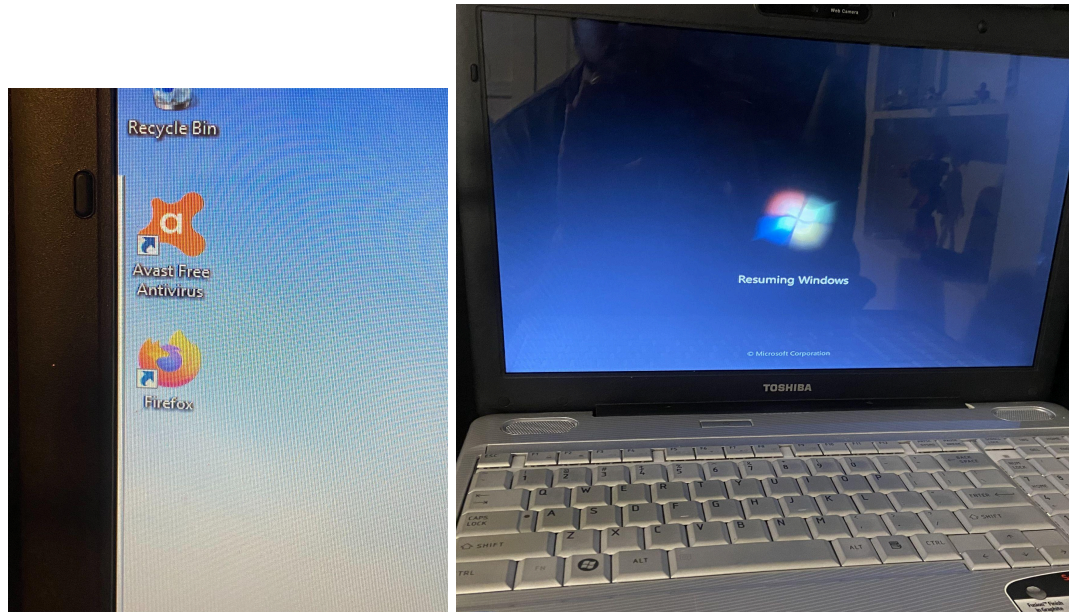
After Norton Antivirus was completely uninstalled, rebooted computer for complete removal. Removing Norton Antivirus reduced lagging by 75%, reinstalled Firefox browser and it is now fully functional with internet access.

- The 2010 Toshiba Windows Laptop device remains functional 100%
- Microsoft Defender is incompatible to Windows 7 laptop, installed Avast Free Antivirus software.

Current Status

The Toshiba Laptop is currently being used and deployed as a test device. Connected the currently functional machine to VLAN20 switchport on my TP-Link Easy switch. It was easily assigned a VLAN20 tag when connected to the configured switchport.

Device is currently used to test on VLAN subinterfaces and IP assignment functionality

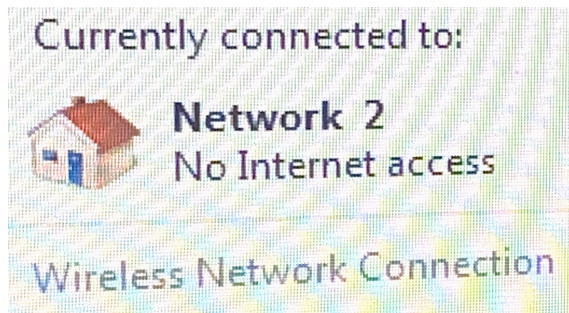
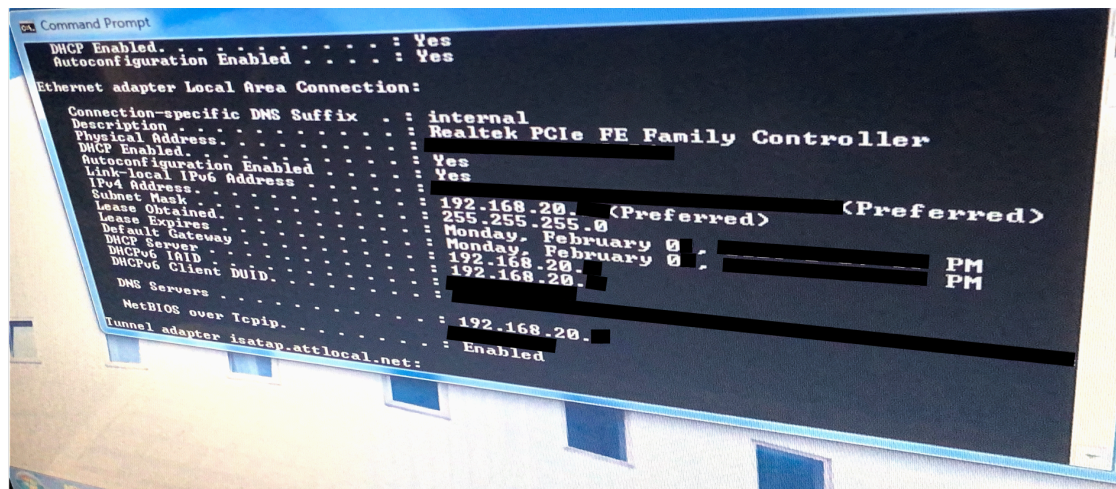


- Status: Good

I connected the Windows 7 Toshiba laptop to VLAN20 switchport on the Layer 2 Tp-Link Easy switch in *Project 2: Proxmox Virtualization & OPNsense Firewall Home Lab Project*.

Pinged the isolated subnet and viewed the laptop is indeed connected to a separate VLAN. Although the IP address is non-routable (no VLAN capabilities on home routers), it serves as a private device connected to a private network. It works best for experimental purposes, I will update soon on its condition.

Results



Windows 7 laptop connected to VLAN IP subnet .20. No internet access since the configured static IP address remains non-routable but remains functional based on OPNsense firewall VLAN configuration and ACL log traffic.

Verdict

2015 X1 Carbon 4th Gen Lenovo Laptop

- X1 Carbon 4th Gen Lenovo Laptop is fully functional, it no longer needs to be constantly charged while turned on. The new laptop battery allows the device to be used without charging constantly.
- The new CMOS battery reduced Time/ Date error configuration on Boot Setup by 100%.
- The laptop is currently operational, fixed and integrated as part of my computer lab.

2010 Windows 7 Toshiba Laptop

- Removed outdated Norton Antivirus Software, reduced lagging by 75%.
- Installed Avast Free Antivirus, 0% malware detected.

- Installed Firefox web browser, replacing outdated Internet Explorer. Toshiba laptop can now access the internet and software applications can now be installed without lagging, lack of connectivity, or bloatware.